Personal Statement — MSc in Intelligent Robotics Engineering, The Hong Kong Polytechnic University

Over the past two years, I have worked on three robots whose control had to be validated on real hardware: a RoboMaster hero robot for which I implemented the gimbal, launcher and chassis control loops; a wheel-legged robot that stays upright while its legs change length; and a three-omni-wheel ballbot that keeps its balance while two arms move on top of it. These projects led me to a clear conclusion: I can implement a controller and make it run, but I now need a more rigorous understanding of why it works and how to design it systematically. That is why I am applying to the MSc in Intelligent Robotics Engineering.

My undergraduate study at Shenzhen Technology University (Computer Science, ranked 6 of 156) gave me a solid base in programming and algorithms. As my robotics projects increasingly required a stronger understanding of control, I also took Principles of Automatic Control and Modern Control Theory. During my internship at the Shenzhen Institute of Artificial Intelligence and Robotics for Society (AIRS), I applied this preparation to a real balance-control problem. I was responsible for the balance control of a dual-arm ballbot. I built and deployed a cascaded PID controller, then modelled the system as a four-state inverted pendulum and used LQR to analyse how state weights trade stability against control effort. The model explained tuning behaviour and actuator saturation that hand tuning alone could not. Compensating arm-induced centre-of-mass shifts with URDF/Pinocchio gravity feedforward showed me how tightly upper-body motion and base balance are coupled. I also integrated the ROS state processing with CAN motor execution and added state initialisation, output limiting and fall-recovery safeguards, so that the loop could be tested repeatedly without damaging the robot. The chassis ultimately maintained dynamic balance while both arms moved along fixed trajectories.

I then extended this control-oriented approach through two laboratory projects, moving from balance control to locomotion and navigation. On a variable-leg-length wheel-legged robot I implemented gain-scheduled LQR and virtual model control with closed-chain leg kinematics, and tuned it on an STM32H7/FDCAN platform. On a car-like mobile robot I built the full chain from EKF/AMCL localisation through Hybrid-A* and TEB planning to MPC tracking, and ran it on the real vehicle. Internships at Future Power and Kaihong then extended my experience in two directions: VR teleoperation of a dual-arm robot, where constrained inverse kinematics keeps the arms tracking the operator without velocity spikes near singularities, and multi-robot platform software on OpenHarmony.

These experiences map directly onto the programme's core. Advanced Control Technology and Robot Motion Planning address the two areas where I have practical results but want theoretical depth. Principles of Robotic Mechanisms will place the closed-chain kinematics I have implemented for one robot within a broader mechanical framework. Embodied Robot Intelligence connects to the agent-based decision chain I built in an OpenHarmony gardening system and to the safety-gated execution chain I am building at Kaihong. I intend to take the dissertation route and use it to produce a properly documented study on balance or navigation control, a type of structured research I have not yet had the opportunity to undertake. The electives in Autonomous Vehicles and Computer Vision and Image Processing fit my navigation and perception experience.

My goal is to work as a robotics algorithm engineer in the Greater Bay Area, on motion control and navigation for mobile and legged platforms, and in time to lead the algorithm side of a robot product. A robotics programme housed in a Department of Mechanical Engineering and focused on robotic systems would give me the structured training in mechanisms and sensing that I lack, while connecting directly with the robotics experience I have built in Shenzhen. For this goal, it offers the most direct next step. This background has given me hands-on hardware experience, a habit of carrying complete systems through validation, and a clear view of the theory I still need to learn.